

# Ars Biologica 2025 - ABIS Summer School

*Permacomputing and Digital Infrastructures*

*Permacomputing a digitální infrastruktury*

*August 21st to 24th 2025, Budweis*



The Permacomputing Summer School, co-organized by the Ars Biologica Independent Study Program (ABIS) for Budweis – European Capital of Culture 2028, the Doctoral Research Department of the Academy of Fine Arts in Prague, and the Uroboros collective, brought together artists, designers, researchers, and students to **engage with computational culture critically**. Over four days, the program employed permaculture principles and permacomputing practices as generative frameworks, focusing on interdependence and contextual sensitivity rather than individual productivity and scale.

The curriculum delved into various aspects of digital infrastructure, treating permacomputing as a **creative lens** for understanding alternatives to current high-abstraction digital technologies and their material and energy dependencies. Discussions challenged fundamental assumptions, such as whether computers should always be online and key themes included **attuning to local contexts and histories, maximizing hardware lifespans, minimizing energy consumption, and emphasizing the reuse and repurposing of existing computational resources**.

Participants explored concepts of **decentralization and local relevance**, focusing on integrating technology with natural contexts, such as solar rhythms. Practical examples such as Low-Tech Magazine's solar-powered website, which operates on a "work when sunny" philosophy and uses static, lightweight designs for energy efficiency, were presented. The summer school also explored speculative projects, such as James Bridle's "Server Farm" and FutureEverything's "Compost Computer," which envision computing infrastructure aging, degrading, or growing like plants.



The program addressed the inherent challenges of the digital landscape, including the **"democratization paradox"** in **AI-powered creative tools**, where access to advanced AI models can create significant inequities in learning outcomes based on financial capacity. It also critically examined concepts like **"enshittification,"** where digital platforms degrade user experience over time to extract more value. The influence of anarchist thought, particularly on self-organization and bottom-up approaches to technology, was a notable thread.

Furthermore, the summer school engaged with **institutional permaculture**, as seen in the Palais de Tokyo's model of treating an institution as a living ecosystem, where invisible processes like governance and communication are as vital as visible outputs like exhibitions, and promoting a **"mindful production"** philosophy of **"doing better rather than doing more,"**.



## Documentations, Notifications, Ramifications, and other Outcomes

- *Digital Infrastructures and Permacomputing Aesthetics NLM*:: a Notebook LM (generative archival system) trained with links, references, papers, notes from chats and discussions, and other materials emerging during the summer



school. References and documents feeding the system are on the left side of the screen, there are notes on the right column, and in the center, a chat interface where to interact with the data that participants generated through the sessions or added through links. Just type your prompts and see what happens ->

<https://notebooklm.google.com/notebook/9ec853b5-d8df-4a7e-a660-5ec9c5720f96>

- *Post-Notifications*: Nádražní 118/6, České Budějovice was the physical location where the summer school took place. A building with a history of cultural dissemination that continues to this day through the work of ECOC2028. It was built in the late 1800s to function as a post office, to share information, and also to hide it, as it became painfully clear when postal workers were tortured for intercepting incriminating evidence during WWII. Echoing the building histories, some summer school participants created a set of post-notifications—alarms, ringtones, pixel stamps, and other sonic artifacts — that both reveal and conceal information emerging from smartphones and other digital devices. They are hidden here, feel free to download and make this secret public:

[https://github.com/EnriqueKi/PermaPrompting/tree/main/\\*\\*PermaComputing\\*\\*](https://github.com/EnriqueKi/PermaPrompting/tree/main/**PermaComputing**)

- *PermaComputingSummerSchool Dithering Tools*: Utilities to batch dither / palette-reduce images and videos in a "low-tech" ordered Bayer style for size reduction and a consistent "Nádražní 118/6 permacomputing aesthetic" Inspired by the work of Kris De Decker and Low Tech Magazine ([https://github.com/lowtechmag/solar\\_v2](https://github.com/lowtechmag/solar_v2)). Find the assets for your permacomputing pleasure here :

[https://github.com/EnriqueKi/PermaPrompting/tree/main/\\*\\*PermaComputing\\*\\*/DitheringTools](https://github.com/EnriqueKi/PermaPrompting/tree/main/**PermaComputing**/DitheringTools)

- *PermaComputingSummerSchool Visuals*: images and videos emerging during the summerschool processed using the SummerSchoolDitheringTools ->

[https://github.com/EnriqueKi/PermaPrompting/tree/main/\\*\\*PermaComputing\\*\\*/Visuals](https://github.com/EnriqueKi/PermaPrompting/tree/main/**PermaComputing**/Visuals)





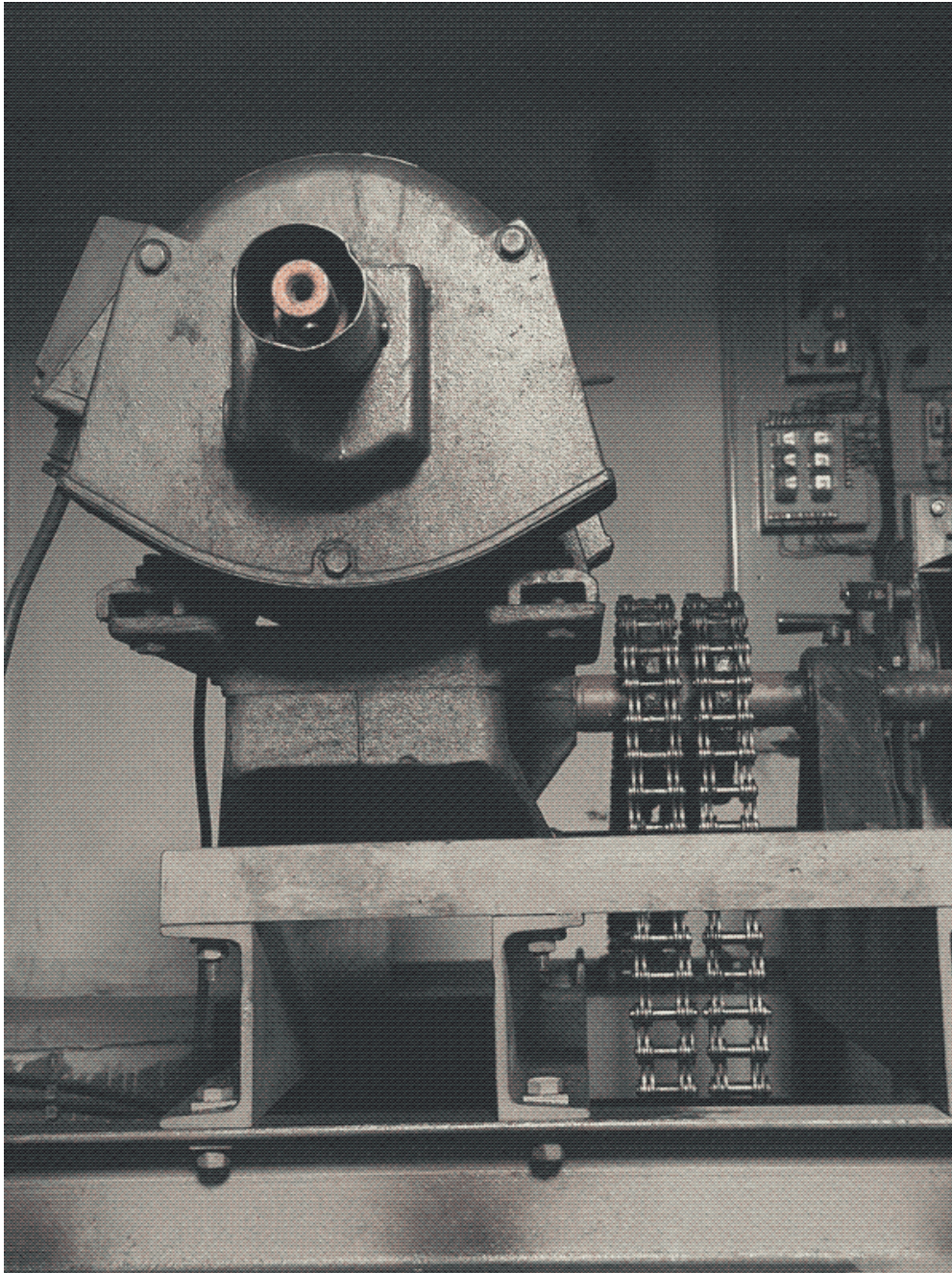
# Full Summer School Program

PROGRAM - <https://pad.monoskop.org/p/AB25-PC-prog>

GARDEN - <https://mapy.com/s/lumugozezo>

<https://www.budejovice2028.eu/udalosti/ars-biologica-letni-skola-2025>

<https://www.budejovice2028.eu/novinky/ars-biologica-2025-umeni-veda-a-budoucnost-krasna>



## Where Do We Meet Next?

- Uroboros Festival -> <https://collective.uroboros.design/>
- Ars Electronica -> <https://ars.electronica.art/panic/en/>
- Transmediale -> <https://transmediale.de/en>
- CTM Festival -> <https://www.ctm-festival.de/>





## Daily Notes and Links

### DAY 1 - 21.8.2025

Cory Doctorow - <https://en.wikipedia.org/wiki/Enshittification>

Ecologies of Technologies pdf -

[https://drive.google.com/file/d/1lXrQAcZLhPTwN\\_eKWMGI4ZCXTEgM573f/view?usp=sharing](https://drive.google.com/file/d/1lXrQAcZLhPTwN_eKWMGI4ZCXTEgM573f/view?usp=sharing)

TBA21 - <https://tba21.org/organismoyearone>

Ocean-Archive.org - <https://ocean-archive.org/>

<https://artycok.tv/cs/post/ocean-as-an-art-space>

Danja - <https://danjavaliev.net/>

NotebookLM - <https://notebooklm.google.com/notebook/9ec853b5-d8df-4a7e-a660-5ec9c5720f96>

### Group presentation

Lukáš

- <https://www.klepsimu.cz/>
- Regenerative design
- Psychedelics - eco-art therapy
- Idea of creative platform to save the world
- Co-organising psychedelic event

Yaryna

- used to work with painting, sound, video
- now experimenting with new media
- working with WayBackMachine - archive.org

Lucie

- Photography - food, fashion
- Artykel - <http://artykel.org/>

Collective Ideation

- Clothes from kombucha
- Permaculture and Architecture

## Kris presentation

- <https://solar.lowtechmagazine.com/2025/06/dressing-and-undressing-the-home/>
- [https://en.wikipedia.org/wiki/Three\\_Horizons](https://en.wikipedia.org/wiki/Three_Horizons)
- <http://joshuacitarella.com/>
- <https://www.regen.network/>

## Feedback session

- top down, bottom up politics
- Stanislav Muller -
- Petr Rokusek -
- catastrophe

## DAY 2 - 22.8.2025

Enrique Encinas - <https://www.aho.no/english/about/employees/enrienci/>

- <https://github.com/EnriqueKi/PermaPrompting/tree/main>
- <https://ah0.no/chat> /// GraciasHelene
- <https://notebooklm.google.com/notebook/9ec853b5-d8df-4a7e-a660-5ec9c5720f96>

<https://www.sense-of-rebellion.com/>

Random links:

- [https://www.reddit.com/r/ChatGPT/comments/1mdn1ar/every\\_shared\\_convo\\_can\\_be\\_viewed\\_by\\_anyone\\_with\\_1/](https://www.reddit.com/r/ChatGPT/comments/1mdn1ar/every_shared_convo_can_be_viewed_by_anyone_with_1/)
- <https://joshuacitarella.com/index.html>
- <https://donotresearch.substack.com/>
- <https://www.instagram.com/bradtroemel/>
- <https://thevoroscope.com/2017/02/24/the-futures-cone-use-and-history/>
- <https://www.versobooks.com/products/3329-medium-hot>
- [https://artinthecageofdigitalreproduction.org/#joel\\_simon](https://artinthecageofdigitalreproduction.org/#joel_simon)
- <https://spikeartmagazine.com/articles/interview-hito-steyerl>
- <https://artreview.com/hito-steyerls-medium-hot-the-age-of-slop-artificial-intelligence-terry-nguyen-opinion/>
- <https://calls.ars.electronica.art/2024/prix/winners/14198/>

Touch designer files

- [https://drive.google.com/drive/folders/14-L1YnokiGv\\_7z6DTX8-fXYF1a18\\_kZl](https://drive.google.com/drive/folders/14-L1YnokiGv_7z6DTX8-fXYF1a18_kZl)
- [https://drive.google.com/drive/folders/1mE52BpfkxMt2\\_3cT1HxU7lk5fwT1NUU?usp=drive\\_link](https://drive.google.com/drive/folders/1mE52BpfkxMt2_3cT1HxU7lk5fwT1NUU?usp=drive_link)
- [https://drive.google.com/drive/folders/11H6e5LUTSkFvUilUCcWFxeL5j1rw3THb?usp=drive\\_link](https://drive.google.com/drive/folders/11H6e5LUTSkFvUilUCcWFxeL5j1rw3THb?usp=drive_link)

Lukas shared

<https://www.greensong.info/natural-senses>

## DAY 3 - 22.8.2025

- PermaPrompt vs. intellectual compost
  - Sense - cards 54 sense, Jojo focus on sensing
  - Participatory - understand the needs of local community
  - Only local
- 
- LowTech Magazine - <https://solar.lowtechmagazine.com/>
  - Fiber Festival - Permacomputing Dossier - <https://www.fiber-space.nl/project/dossier-permacomputing/>
  - [https://monoskop.org/Permacomputing\\_Aesthetics](https://monoskop.org/Permacomputing_Aesthetics)
  - <https://monoskop.org/Permacomputing>
  - <https://github.com/EnriqueKi/PermaPrompting>
  - <https://permacomputing.net>
  - <https://waag.org/en/project/designing-regenerative-technologies/>
  - <https://conference.publicspaces.net/en/session/regenerative-technologies-and-permacomputing@>
  - <https://opencollective.com/node9>
  - <https://futureeverything.org/portfolio/entry/compost-computer/>
  - <https://jamesbridle.com/works/server-farm>
  - <https://artreview.com/power-100/>
  - Budweis 2028 - BidBook - <https://www.budejovice2028.eu/files/bidbook-w>